



Sir Syed University of Engineering & Technology
Faculty of Computing and Applied Sciences
Department of Computer Science & Information Technology

Mid Term Examinations (Fall 2022)

Course Code with Title	CS-211: Discrete Structure	Program	Bachelor of Computer Science BS (CS)
Instructors	Fariha Ansari/M. Etezaz Ibrahim	Semester	3 rd
Start date & Time	Dec 06, 2022 at 11:30 AM	End Time	01:00 PM
Maximum Marks	30		

IMPORTANT INSTRUCTIONS:

- Attempt All Questions.

Question 1

(05+05)

(a). Let p be "Jupiter is a planet" and q be "Pakistan is an Island" Express in Simple sentence the meaning of following mathematical propositions.

(i) $\neg p$ (ii) $p \wedge \neg q$ (iii) $\neg p \vee q$ (iv) $p \leftrightarrow q$

(b). Show that $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

Question 2

(02+03+02+03)

(a). How many students must be in a class to guaranty that atleast two students receive the same score on the final exam, if the exam is graded on a scale from 0 to 50? **56**

(b). Suppose that f is defined recursively by $f(n) = \begin{cases} 3; & n = 0 \\ 2f(n-1) + 3; & n > 0 \end{cases}$
 Find $f(1)$, $f(2)$, $f(3)$

3, 2*3+3, 4*3+3

(c). Apply Floor and Ceiling rule to calculate the given expression

$\left\lceil \frac{1}{4} + \left\lfloor \frac{3}{2} + \frac{1}{2} \right\rfloor + \left\lfloor \frac{3}{2} + \frac{1}{6} \right\rfloor \right\rceil = 5$

(d). Consider the function from the set of students in a class. Under what conditions is the function one to one; if it assign to a student his/her

- CNIC number
 - Roll number
 - Resident Area
- Explain with reason.



Sir Syed University of Engineering & Technology
Faculty of Computing and Applied Sciences
Department of Computer Science & Information Technology

Question 3

(04+04+02)

(a). Let $R = \{(1,1), (2,2), (3,3), (4,4), (2,1), (3,1), (3,2), (4,1), (4,2), (4,3)\}$. Express R in the form of Directed Graph. Also determine if R is a POSET or not?

(b). Determine whether the relation represented by is an Equivalence Relation? *not*

	a	b	c
1	1	1	1
2	0	1	1
3	1	1	1

(c). Among the first 1000 positive integers, determine the integers which are not divisible by 4 nor by 5. *600*